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RESEARCH ARTICLE

Taste Difference Thresholds for Monosodium Glutamate and Sodium Chloride in Pigtail Macaques (*Macaca nemestrina*) and Spider Monkeys (*Ateles geoffroyi*)

MATTHIAS LASKA[1]*, JOSEFIN PERSSON SUORRA[1], ROSA MARIELA RIVAS BAUTISTA[2], AND LAURA TERESA HERNANDEZ SALAZAR[2]
[1]*IFM Biology, Linköping University, Linköping, Sweden*
[2]*Instituto de Neuro-Etologia, Universidad Veracruzana, Xalapa, Veracruz, Mexico*

The purpose of this study was to determine taste difference thresholds for monosodium glutamate (MSG) and sodium chloride (NaCl) in pigtail macaques (*Macaca nemestrina*) and spider monkeys (*Ateles geoffroyi*). Using a two-bottle preference test of brief duration, three animals of each species were presented with four different reference concentrations of 50, 100, 200, and 400 mM of a tastant and tested for their ability to discriminate these from lower concentrations of the same tastant. The just noticeable differences (JNDs), expressed as Weber ratios ($\Delta I/I$), were found to range from 0.1 to 0.5 for MSG and 0.2 to 0.45 for NaCl in the pigtail macaques, with a significant tendency for higher Weber ratios with higher reference concentrations. In the spider monkeys, JNDs ranged from 0.15 to 0.4 for MSG and 0.1 to 0.25 for NaCl, with Weber ratios staying fairly constant across the reference concentrations tested. Thus, the JNDs were found to be generally similar in both species and to be at least as low as those found in humans for MSG and NaCl, as well as those found in spider monkeys for sucrose. The results support the assumption that both pigtail macaques and spider monkeys may use differences in perceived intensity of MSG and NaCl as a criterion for food selection. Am. J. Primatol. 70:839–847, 2008. © 2008 Wiley-Liss, Inc.

Key words: taste difference thresholds; monosodium glutamate; sodium chloride; pigtail macaques; spider monkeys

INTRODUCTION

In nonhuman primates, the sense of taste has been investigated anatomically [e.g. Emura et al., 2002; Rolls & Scott, 2003], behaviorally [e.g. Laska et al., 2007; Visalberghi et al., 2003], and electrophysiologically [e.g. Danilova et al., 2002; Pritchard et al., 2005] in a number of species. With regard to sensitivity of this sensory system most studies have so far concentrated on determination of detection thresholds [e.g. Glaser, 1986; Laska et al., 2000a], whereas only little information is available on the ability of nonhuman primates to detect intensity differences between taste stimuli [Laska et al., 1999a; Wagner et al., 1965]. However, apart from a sufficiently high absolute taste sensitivity enabling the detection of tastants indicating the nutritional value of a potential food, a well-developed relative taste sensitivity, that is the ability to discriminate between different concentrations of a tastant, may also be important; for example, to optimize energy yield or to avoid food containing too high concentrations of toxic compounds. Further, the food selection behavior of primates suggests that they may indeed use perceived differences in the concentration of certain taste substances as a criterion of food palatability [Hladik & Simmen, 1996]. This supposition is also supported by the few reports of taste difference thresholds in nonhuman primates, which showed, for example, spider monkeys to have a highly developed ability to detect small differences between suprathreshold concentrations of sucrose. With this disaccharide, the spider monkeys displayed just noticeable differences (JNDs) of 0.075–0.25 when expressed as Weber ratios ($\Delta I/I$) [Laska et al., 1999a]. This should not be surprising given the importance of sweet-tasting carbohydrates for food choice in highly frugivorous primates such as the spider monkey [Laska et al., 2000b].

Monosodium glutamate (MSG) and sodium chloride (NaCl) are the prototypic representatives of the taste qualities ''umami'' and ''salty.'' The taste of MSG is believed to indicate the presence of animal

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*Correspondence to: Matthias Laska, IFM Biology, Linköping University, SE-581 83 Linköping, Sweden.
E-mail: malas@ifm.liu.se

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protein [Ninomiya, 1998] and the taste of NaCl has been found to be attractive for many species as a proper supply with this salt is crucial for maintenance of homeostasis [McCaughey & Scott, 1998].

Spider monkeys have been reported to be highly sensitive for both tastants and displayed preference thresholds of 2 mM for MSG and 1 mM for NaCl. Pigtail macaques, in contrast, that were tested in parallel needed concentrations of 50 mM MSG and 20 mM NaCl to display a marked preference for these tastants when tested against water [Laska & Hernandez Salazar, 2004]. This raises the question whether the marked difference in absolute taste sensitivity for MSG and NaCl between these two primate species is reflected by similar differences in their relative taste sensitivity, that is their ability to detect small differences in suprathreshold concentrations.

It was therefore the aim of this study to determine taste difference thresholds in spider monkeys and pigtail macaques. To this end, we employed two-bottle preference tests of brief duration (1 min). This method makes it possible to directly measure preferences for or aversions to tastants and largely rules out the influence of postingestive factors on the animal's ingestive behavior. More specifically, the monkeys were given a choice between fixed reference concentrations of either MSG or NaCl and systematically varying concentrations of the same tastant, thus allowing us to determine a taste difference threshold—as a first and conservative measure of the animals' differential sensitivity.

MATERIALS AND METHODS

Animals

Testing was carried out using two males and one female adult pigtail macaque (*Macaca nemestrina*), and three female adult spider monkeys (*Ateles geoffroyi*). The rationale for using these primate species was that data on taste preference thresholds for NaCl and MSG are at hand [Laska & Hernandez Salazar, 2004] allowing us to assess possible correlations between absolute and relative sensitivity for these tastants. Additionally, data on taste difference thresholds for sucrose in the spider monkey have been published [Laska et al., 1999a] allowing us to compare relative sensitivity between tastants within this species.

Animals of both species were housed as social groups in enclosures ($10 \times 6 \times 4$ m for the spider monkeys, and $6 \times 5 \times 3$ m for the pigtail macaques) with adjacent single cages that could be closed by sliding doors to allow temporary separation of animals for individual testing (for details of maintenance, see Laska & Hernandez Salazar, 2004; Laska et al., 2003). Animals were fed commercial monkey chow, fresh fruit, and vegetables ad libitum. The amount of food offered daily to the animals was such that left-overs were still present on the floor the next morning and thus it was unlikely that ravenous appetite affected the animals' ingestive behavior in the tests. As spider monkeys do not normally drink from open sources but meet their water requirements by consuming juicy fruits, no water deprivation schedule was adopted. The pigtail macaques, in contrast, need to drink from open water sources to meet their water requirements and were deprived of water overnight before testing on the following morning to ensure cooperation. As a consequence of their lack of water deprivation, the spider monkeys only cooperate in two-bottle preference tests as long as at least one of the alternatives presented to them is perceived as attractive. In this study, the two highest reference concentrations of NaCl presented, that is, 200 and 400 mM, respectively, were clearly not attractive to the animals that thus did not cooperate in these two test series.

All animals had served in previous studies using the same method [Laska & Hernandez Salazar, 2004; Laska et al., 2003], they were trained to enter the single cages voluntarily, and were completely accustomed to the procedure described below.

Procedure

Taste difference thresholds for MSG and NaCl (reagent grade, Merck, Darmstadt, Germany) were determined using a two-bottle preference test of short duration [Richter & Campbell, 1940]. Three times each day, between 9 a.m. and 11 a.m., and approximately 2 hr, 1 hr, and shortly before feeding, the animals were separated and allowed 1 min to drink from a pair of simultaneously presented graduated cylinders with metal drinking spouts.

Difference threshold values or JNDs were established at four reference concentrations of 50, 100, 200, and 400 mM of a taste substance, by pairing each of these with lower concentrations of the same taste substance (e.g. the 100 mM reference versus 10, 20, 50, 60, 70, 80, and 90 mM) until the animals failed to show a significant preference for one of the two alternatives.

Each pair of stimuli was presented 10 times, and the position of the stimuli was randomized in order to counterbalance possible position preferences. Care was taken that the animals could not gather information as to the content of the cylinders by attaching them on the outside of the mesh of a test cage in such a way that only the metal drinking tubes were visible, and between trials the cylinders were removed.

To keep up the animals' motivation and willingness to cooperate, testing did not follow a strict ascending staircase procedure but followed a pseudo-randomized scheme in which trials with large and thus presumably readily discriminable concentration differences of a given taste substance were alternated with trials employing small and thus presumably less easily discriminable concentration differences.

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The experiments reported here comply with the *Guide for the Care and Use of Laboratory Animals* (National Institutes of Health Publication no. 86-23, revised 1985) and also with current German and Mexican laws.

Data Analysis

For each animal, the amount of liquid consumed from each bottle was recorded, summed for the ten test trials with a given stimulus combination, converted to percentages (relative to the total amount of liquid consumed from both bottles), and 66.7% (i.e. 2/3 of the total amount of liquid consumed) was taken as the criterion of preference. This criterion was chosen for reasons of comparability of data as the same criterion had been used in previous studies using the same method with the same or other primate species [Laska, 1994, 1996, 1997, 1999, 2000; Laska & Hernandez Salazar, 2004; Laska et al., 1996, 1998, 1999a,b, 2000b, 2001, 2003], and to avoid misinterpretation of data because of a too liberal criterion.

Additionally, two-tailed binomial tests were performed and an animal was only regarded as significantly preferring one of the two alternative stimuli if it reached the criterion of 66.7%, and consumed more from the bottle containing the preferred stimulus in at least eight out of ten trials (binomial test, $P<0.05$).

Preliminary analysis of the data indicated that there were no reliable differences in choice behavior and liquid consumption between the first, second, and third presentation of the day. Intraindividual variability in the amount of liquid consumed across the ten test trials with a given stimulus combination was low and averaged less than 20%. Thus, a theoretically possible bias in the overall preference score because of excessive drinking in aberrant trials did not occur. The data for each individual animal obtained in the ten test trials with a given stimulus combination are therefore reported as mean percentage values.

RESULTS

Pigtail Macaques

Figure 1 summarizes the performance of the pigtail macaques in discriminating between different concentrations of MSG. All three animals signifi-

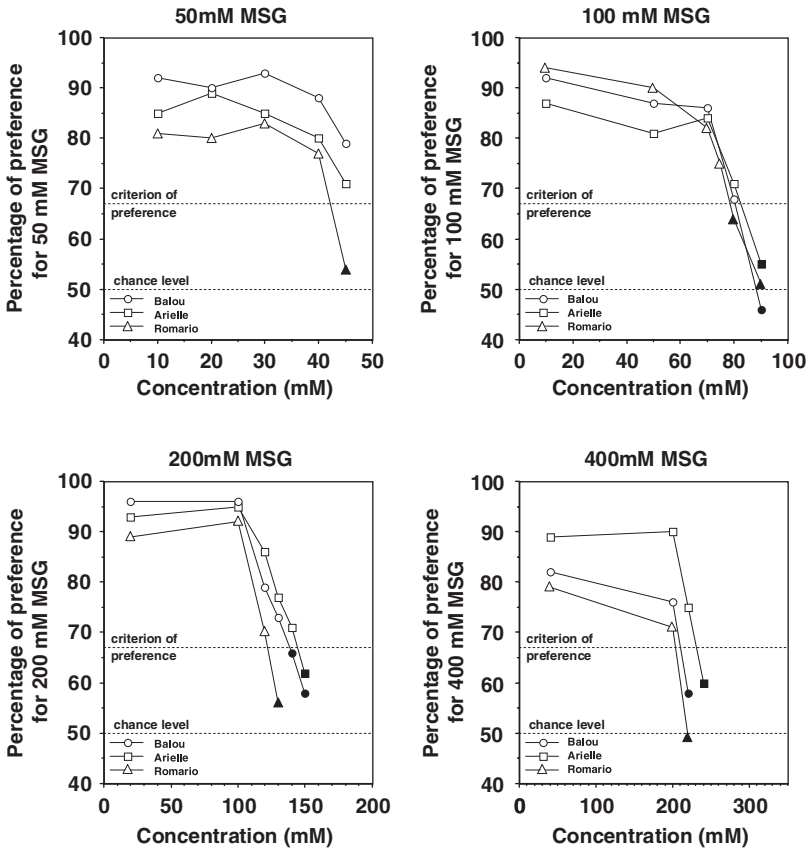


Fig. 1. Performance of the pigtail macaques in discriminating between different concentrations of monosodium glutamate (MSG). Each data point represents the mean value of ten test trials of 1 min per animal and stimulus combination. Filled symbols indicate stimulus combinations that were not discriminated significantly above the chance level (binomial test, $P>0.05$). Dotted lines indicate the chance level and criterion of preference, respectively (binomial test, $P<0.05$).

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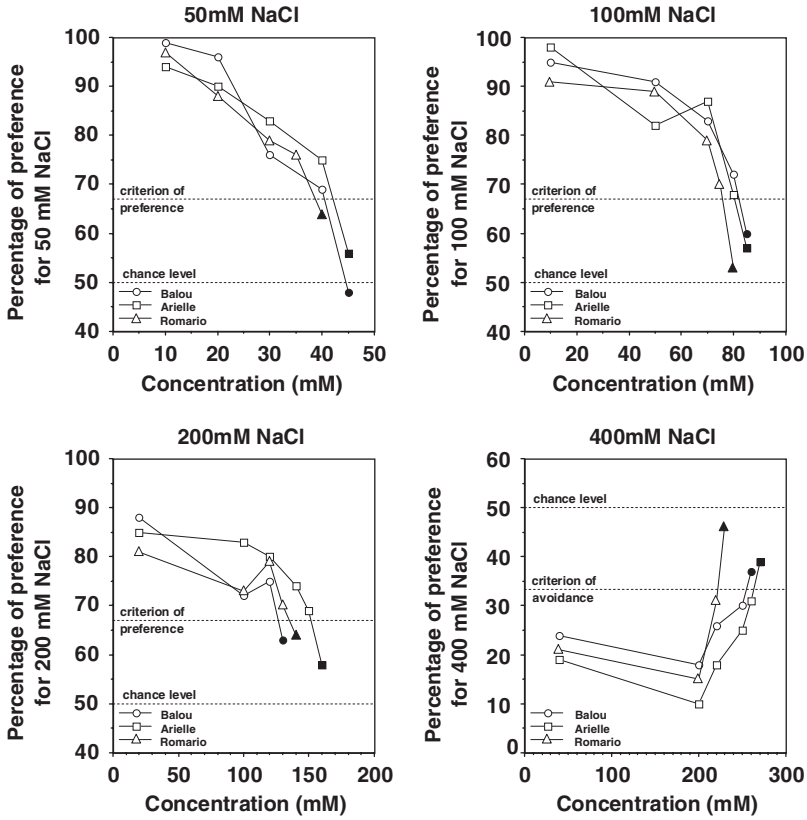


Fig. 2. Performance of the pigtail macaques in discriminating between different concentrations of sodium chloride (NaCl). Each data point represents the mean value of ten test trials of 1 min per animal and stimulus combination. Filled symbols indicate stimulus combinations that were not discriminated significantly above the chance level (binomial test, $P > 0.05$). Dotted lines indicate the chance level, criterion of preference, and criterion of avoidance, respectively (binomial test, $P < 0.05$).

cantly discriminated the reference concentration of 50 mM from a 40 mM alternative, and two of the animals even mastered the discrimination between 50 and 45 mM MSG (see the upper left panel). At a concentration of 100 mM two animals succeeded in distinguishing this reference from 80 mM MSG, whereas one animal only showed a significant preference for the reference against 75 mM MSG (see the upper right panel). All three animals significantly discriminated the reference concentration of 200 mM MSG from a 120 mM alternative, and one animal each even mastered the discrimination between this reference and 130 and 140 mM, respectively (see the lower left panel). At a concentration of 400 mM all three animals succeeded in discriminating this reference from 200 mM MSG, and one animal even distinguished this reference from 220 mM (see the lower right panel).

Figure 2 shows the performance of the pigtail macaques in discriminating between different concentrations of NaCl. Two of the animals significantly discriminated the reference concentration of 50 mM from a 40 mM alternative, whereas one animal only showed a significant preference for the reference against 35 mM NaCl (see the upper left panel). At a concentration of 100 mM two animals succeeded in distinguishing this reference from 80 mM NaCl and one animal from 75 mM (see the upper right panel). All three animals significantly discriminated the reference concentration of 200 mM from 120 mM NaCl, and one animal each mastered the discrimination between this reference and 130 and 150 mM, respectively (see the lower left panel). At a concentration of 400 mM all three animals significantly preferred a lower concentration of 200 mM NaCl over the reference, and one animal each succeeded in distinguishing this highest reference from 220, 250, and 260 mM, respectively (see the lower right panel).

Spider Monkeys

Figure 3 summarizes the performance of the spider monkeys in discriminating between different concentrations of MSG. Two of the animals significantly discriminated the reference concentration of 50 mM from a 40 mM alternative, whereas one animal only showed a significant preference for the reference against 30 mM MSG (see the upper left panel). At a concentration of 100 mM all three animals succeeded in distinguishing this reference from 60 mM MSG and one animal each even showed a significant preference for the reference against 65

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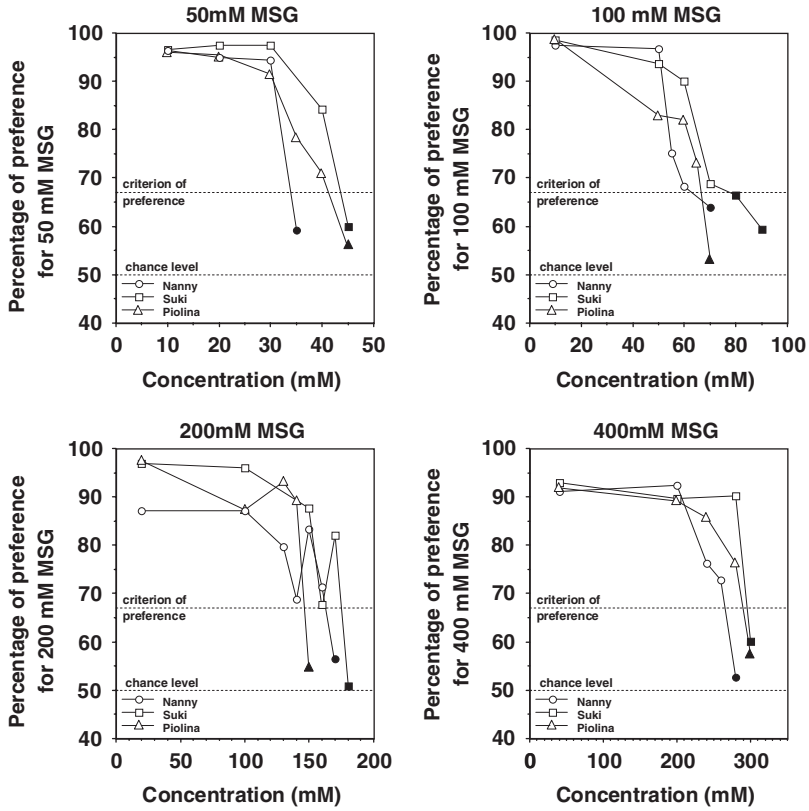


Fig. 3. Performance of the spider monkeys in discriminating between different concentrations of monosodium glutamate (MSG). Each data point represents the mean value of ten test trials of 1 min per animal and stimulus combination. Filled symbols indicate stimulus combinations that were not discriminated significantly above the chance level (binomial test, $P > 0.05$). Dotted lines indicate the chance level and criterion of preference, respectively (binomial test, $P < 0.05$).

and 70 mM MSG, respectively (see the upper right panel). When presented with a concentration of 200 mM MSG one animal each mastered to discriminate this reference from a 140, 160, and 170 mM alternative, respectively (see the lower left panel). At a concentration of 400 mM, two animals succeeded in discriminating this reference from 280 mM MSG, whereas one animal only distinguished this reference from 260 mM (see the lower right panel).

Figure 4 shows the performance of the spider monkeys in discriminating between different concentrations of NaCl. All three animals significantly discriminated the reference concentration of 50 mM from a 40 mM alternative and one animal even mastered the discrimination between 50 and 45 mM NaCl (see the left panel). At a concentration of 100 mM one animal each succeeded in distinguishing this reference from 75, 80, and 90 mM NaCl (see the right panel). Attempts to test the spider monkeys with reference concentrations of 200 and 400 mM NaCl failed because of lack of cooperation.

Intra- and Interspecific Comparison

Figure 5 summarizes the taste difference thresholds for MSG and NaCl determined in the pigtail macaques and the spider monkeys, expressed as Weber ratios ($\Delta I/I$). For the pigtail macaques, Weber ratios increased monotonously with increasing reference concentration (from 0.1–0.2 at 50 mM to 0.45–0.5 at 400 mM, see the upper left panel). Accordingly, a highly significant positive correlation between reference concentration and Weber ratios was found (Spearman $r_s = 0.96$, $P < 0.002$). A similar trend, although not as clearcut, was found with NaCl in this species (from 0.2–0.3 at 50 mM to 0.35–0.45 at 400 mM, see the lower left panel). Here, too, a significant positive correlation was found (Spearman $r_s = 0.77$, $P < 0.02$). In the spider monkeys, in contrast, Weber ratios for MSG stayed fairly constant across the reference concentrations tested (0.2–0.4 at 50 mM to 0.3–0.35 at 400 mM, see the upper right panel). Accordingly, no significant correlation between the two measures was found (Spearman $r_s = 0.10$, $P > 0.05$). The same is true for the Weber ratios found with NaCl in this species (0.1–0.2 at 50 mM and 0.1–0.25 at 100 mM, see the lower right panel).

DISCUSSION

The results of this study can be regarded as a first and conservative approximation of the differential taste sensitivity of pigtail macaques and spider

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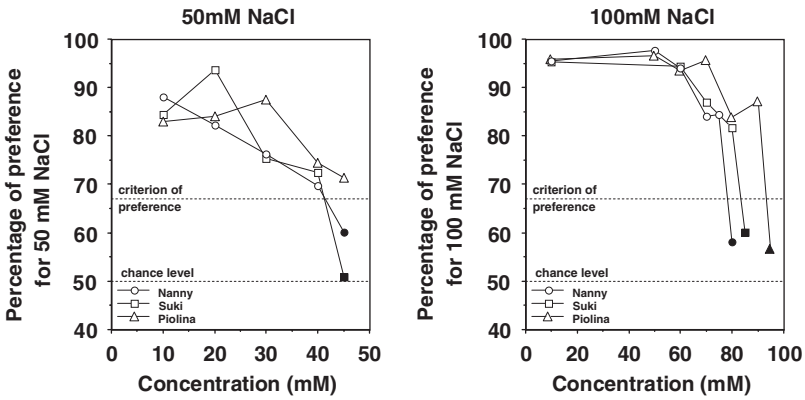


Fig. 4. Performance of the spider monkeys in discriminating between different concentrations of sodium chloride (NaCl). Each data point represents the mean value of ten test trials of 1 min per animal and stimulus combination. Filled symbols indicate stimulus combinations that were not discriminated significantly above the chance level (binomial test, $P > 0.05$). Dotted lines indicate the chance level and criterion of preference, respectively (binomial test, $P < 0.05$).

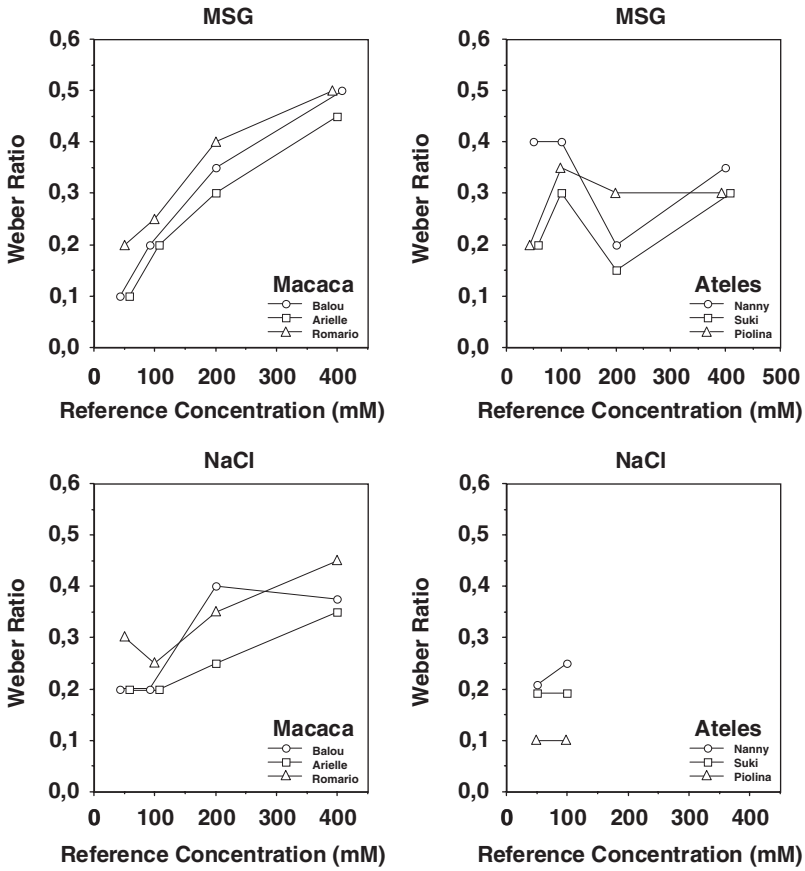


Fig. 5. Taste difference thresholds at four different reference concentrations of monosodium glutamate (MSG, upper panels) and sodium chloride (NaCl, lower panels) in pigtail macaques (*Macaca*, left panels) and spider monkeys (*Ateles*, right panels), expressed as Weber ratios ($\Delta I/I$). Each data point represents an individual animal.

monkeys for MSG and NaCl. Although only three animals per species were tested, the results appear robust as interindividual variability was low and generally in the range reported in studies on human taste difference thresholds [Pfaffman et al., 1971]. In fact, for the majority of cases the range of Weber ratios at a given reference concentration between the highest- and the lowest-scoring animal of a species was 0.1 or even smaller (see Fig. 5). Further, with both substances tested, the animals' performance with the smallest concentration differences presented usually dropped to a chance level, suggesting

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TABLE I. Weber Ratios for MSG and NaCl in Humans and Nonhuman Species

Species	Weber ratio	Substance	Source
Pigtail macaque	0.1–0.5	MSG	This study
Spider monkey	0.15–0.4	MSG	This study
Human	0.26	MSG	Mojet et al. [2003]
Human	0.14–0.28	MSG	Yamaguchi [1967]
Pigtail macaque	0.2–0.45	NaCl	This study
Spider monkey	0.1–0.25	NaCl	This study
Human	0.117–0.133	NaCl	Laurila et al. [1996]
Human	0.072–0.087	NaCl	Johansson et al. [1973]
Human	0.20–0.24	NaCl	Nordin et al. [2003]
Human	0.112–0.168	NaCl	Laing et al. [1993]
Rat	0.17–0.18	NaCl	Brosvic and Slotnick [1986]
Rat	0.49	NaCl	Colbert et al. [2004]
Harbour seal	0.05–0.20	NaCl	Sticken and Dehnhardt [2000]

MSG, monosodium glutamate; NaCl, sodium chloride.

that the statistically significant discrimination between larger concentration differences of a given taste substance was indeed based on taste perception and not on other cues.

Table I compares the performance of both species tested here with the few data available on Weber ratios for MSG and NaCl in humans and nonhuman species.

Our finding that the taste difference thresholds found with the pigtail macaques and the spider monkeys were at least in the same range as, and in some cases even lower than, those established in humans is remarkable considering that the sophisticated psychophysical signal detection methods employed with human subjects are likely to be more sensitive compared with the simple two-bottle preference test used with the monkeys [Spector, 2003]. This supposition is also supported by findings from studies in other nonhuman species that showed taste thresholds based on simple preference procedures to be higher compared with thresholds obtained with operant conditioning procedures [Reilly et al., 1994] or with taste-aversion conditioning procedures [Harder et al., 1989]. Taken together, this suggests that the differential taste sensitivity for MSG and NaCl in both primate species tested here might not only match but perhaps even exceed that of *Homo sapiens*.

A comparison with the two only other nonhuman species tested so far shows rats to have Weber ratios for NaCl that are similar to those found in this study, whereas harbour seals display the lowest value ever published (see Table I). This raises the question as to possible explanations for such differences in performance between species and for the different patterns of performance found in this study (see Fig. 5). The high gustatory resolving power for differences in salinity found in harbour seals is discussed as an evolutionary adaptation allowing chemosensory orientation in the marine environment [Sticken & Dehnhardt, 2000]. In nonhuman primates, differences in dietary habits have repeatedly been shown to plausibly explain differences in taste performance. Among New World primates, for example, the degree of frugivory has been found to correlate positively with sensitivity for soluble carbohydrates [Laska, 1996; Laska et al., 1996]. This should not be surprising given the importance of food-associated mono- and disaccharides as a primary source of metabolic energy in frugivorous species. Similarly, taste thresholds for MSG and NaCl have been reported to correlate negatively with the proportion of animal matter in the diet of primates [Laska & Hernandez Salazar, 2004]. This finding, too, appears to make sense considering that MSG is believed to indicate the presence of dietary animal protein and thus should be particularly important for species feeding on a low-protein diet [Ninomiya, 1998].

The results of this study, however, showed pigtail macaques and spider monkeys to display the same range of relative sensitivity in terms of Weber ratios for MSG and NaCl, whereas they differ markedly in their absolute sensitivity for these tastants [Laska & Hernandez Salazar, 2004] and despite the fact that they differ considerably in their intake of animal matter, which has been reported to be $\leq 1\%$ in the spider monkey compared with $\geq 13\%$ in the pigtail macaque [Caldecott, 1986; Kinzey, 1997]. A closer look at the Weber ratios for MSG in these two primate species, however, reveals that the pigtail macaques tend to display lower values than the spider monkeys at the two lowest reference concentrations tested and a reverse trend with the two highest reference concentrations (see Fig. 5, upper panels). The lowest reference concentration tested with both tastants, that is 50 mM, already represents the upper limit of MSG and NaCl concentrations that are usually found in the natural diet of nonhuman primates [Committee on Animal Nutrition, 2003]. Thus, the higher reference concentrations tested here may be considered as

unphysiological, at least with regard to their occurrence in the natural diet of the primate species under investigation. This interpretation is also supported by our findings that the spider monkeys failed to cooperate when presented with the two highest reference concentrations of NaCl and that the pigtail macaques rejected the highest reference concentration of NaCl (see Fig. 2).

Our finding of a higher differential sensitivity for MSG in the pigtail macaques than the spider monkeys at the physiological concentration of 50 mM might therefore be interpreted in terms of optimal foraging theory [Campbell, 1987]. Pigtail macaques include a higher proportion of animal matter into their diet than spider monkeys and thus may gain considerably in protein yield if their ability to detect intensity differences for MSG is highest with low reference concentrations, whereas spider monkeys do not. This hypothesis, however, needs to be tested further by including a larger variety of primate species into future studies on taste difference thresholds.

Taken together, the results of this study support the assumption that both pigtail macaques and spider monkeys may use differences in perceived intensity of MSG and NaCl as a criterion for food selection.

ACKNOWLEDGMENTS

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